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TOE–601

B. TECH. (ECE) (SIXTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

PROGRAMMING USING PYTHON

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the different features, applications, advantages and disadvantages of Python ? (CO1)

OR

(b) "Python has developed as an open source project." Justify this statement. (CO1)

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2. (a) Write short notes on list, tuple, set, dictionary and string in Python. (CO1)

OR

(b) With the help of an example, explain the different types of arithmetic operators in Python. (CO1)

3. (a) Write a program to calculate salary of an employee given his basic pay (to be entered by the user), HRA = 10 per cent of the basic salary, TA=5 per cent of basic pay. Define HRA and TA as constants and use them to calculate the salary of an employee. (CO1)

OR

(b) Explain conditional branching statements supported by Python. (CO2)

4. (a) Explain the utility of break and continue statement with the help of an example. (CO2)

OR

(b) (i) Write a Python program to determine whether a person is eligible to vote or not. If he or she is not eligible, display how many years are left to be eligible ?

(ii) Write a Python program to test whether a number entered by the user is positive, negative or equal to zero. (CO2)

5. (a) Write short notes on the following : (CO2)

(i) Keyword arguments and Default arguments

(ii) Lambda function

OR

(b) (i) Write a program that uses lambda function to multiply two numbers.

(ii) Write a program that finds the greatest of three numbers using functions. Pass the numbers as arguments. (CO2)